

Exploitative and Hierarchical Antagonism in a Cooperative Bacterium

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Social organisms that cooperate with some members of their own species, such as close relatives, may fail to cooperate with other genotypes of the same species. Such noncooperation may take the form of outright antagonism or social exploitation. *Myxococcus xanthus* is a highly social prokaryote that cooperatively develops into spore-bearing, multicellular fruiting bodies in response to starvation. Here we have characterized the nature of social interactions among nine developmentally proficient strains of *M. xanthus* isolated from spatially distant locations. Strains were competed against one another in all possible pairwise combinations during starvation-induced development. In most pairings, at least one competitor exhibited strong antagonism toward its partner and a majority of mixes showed bidirectional antagonism that decreased total spore production, even to the point of driving whole populations to extinction. Differential response to mixing was the primary determinant of competitive superiority rather than the sporulation efficiencies of unmixed populations. In some competitive pairings, the dominant partner sporulated more efficiently in mixed populations than in clonal isolation. This finding represents a novel form of exploitation in bacteria carried out by socially competent genotypes and is the first documentation of social exploitation among natural bacterial isolates. Patterns of antagonistic superiority among these strains form a highly linear dominance hierarchy. At least some competition pairs construct chimeric, rather than segregated, fruiting bodies. The cooperative prokaryote *M. xanthus* has diverged into a large number of distinct social types that cooperate with clone-mates but exhibit intense antagonism toward distinct social types of the same species. Most lengthy migration events in nature may thus result in strong antagonism between migratory and resident populations, and this antagonism may have large effects on local population sizes and dynamics. Intense mutual antagonism appears to be more prevalent in this prokaryotic social species than has been observed in the eukaryotic social slime mold *Dictyostelium discoideum*, which also exhibits multicellular development. The finding of several cases of facultative social exploitation among these natural isolates suggests that such exploitation may occur frequently in nature in many prokaryotes with cooperative traits.